

PROJECT EXECUTIVE SUMMARY

Title: Exascale Simulations of Excited-State Phenomena in Quantum Materials

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Applying Institution/Organization: University of Southern California

Resource Name(s) and Number of Node Hours Requested: Frontier 450,000 Node Hours, Aurora 400,000 Node Hours, Polaris 100,000 Node Hours

Amount of Storage Requested: File system storage of 150 TB; Archival storage of 50 TB

Executive Summary:

The overarching goal of this project is to apply and advance the state-of-the-art *ab initio* many-body Green's function approaches for understanding and predicting excited-state phenomena in quantum materials. Multiparticle interactions such as electron-electron, electron-hole, electron-phonon couplings drive many exotic quantum phases and orders (such as superconductivity and excitonic insulators) and are essential in excited-state scattering and quantum decoherence processes. Accurate first-principles descriptions of these many-body interactions and excited-state phenomena pose grand challenges in theoretical formulation and software implementation, and have high demands for massive computational resources. Through decades of developments, the GW and GW-Bethe-Salpeter equation (GW-BSE) approaches have been established as the standard first-principles methods for studying quasiparticle excitations, optical properties, and excitons in a broad range of materials. More recently, new advancements in GW perturbation theory (GWPT) and time-dependent GW (TDGW) approaches have further enriched the many-body Green's function framework to capture correlated electron-phonon coupling and nonequilibrium dynamics of excited states, respectively.

The proposed research focuses on quasiparticle excitations, optical properties, time-dependent and nonequilibrium phenomena, multiparticle excitations, and coupling between phonons and correlated electrons and excitons. The combination of GW, GW-BSE, GWPT, and TDGW approaches offers unique opportunities to reveal emergent correlated and coherent excited-states phenomena in quantum materials that are previously inaccessible due to the lack of both appropriate advanced methodologies and the needed scale of computational resources. We aim to gain deep understanding, at the many-electron interacting level, of (1) how phonons couple to correlated electrons and excitons and hence shape the single-photon emission, photocatalytic pathways, and the formation of self-trapped excitons, (2) how excitons arise in strongly correlated Wigner-crystal states in moiré superlattices, and condense into excitonic insulator phases in two-dimensional materials, (3) how excited states evolve in nonequilibrium and how they can be engineered through external stimuli such as optical pump. The pursuits of these scientific quests are strongly supported by (4) the exascale capability enabled in our software advancements.

The central software package in this proposal is BerkeleyGW, which is at a unique position in the research of excited-state phenomena. First, BerkeleyGW has pioneered and advanced the development of many state-of-the-art first-principles methodologies, including GW, GW-BSE, GWPT, and TDGW. Therefore, the excited-state phenomena can be treated consistently at the same many-electron level. Second, BerkeleyGW has enabled excellent GPU acceleration at scale (Finalist for Gordon Bell Prize 2020) and true portability across all major GPU architectures, namely NVIDIA, AMD, and Intel. Very recently, BerkeleyGW has achieved over 500 PetaFLOP/s of sustained performance (~30–40% of peak) on both exascale supercomputers Frontier at OLCF (full scale) and Aurora at ALCF (87.5% of full scale), handling materials system sizes up to ~10,000 atoms.

Our team is formed by six research groups led by four junior researchers and two senior researchers. The team members have demonstrated leadership and expertise in developing and applying novel many-body methodologies with high-performance computing resources. The six groups of students, postdoctoral scholars, and PIs will work synergistically with complementary expertise to advance the many-body descriptions of excited-state phenomena in quantum materials with exascale capability.

PROJECT NARRATIVE

1. SIGNIFICANCE OF RESEARCH

Excited-state phenomena in materials typically give rise to their defining attributes and determine their usefulness. Computational discovery and design of functional materials utilizing excited-state properties require accurate and predictive descriptions of many-body interactions, such as electron-electron (e - e), electron-hole (e - h), electron-phonon (e -ph) couplings, and the correlation and coherence of excitations, both in- and out-of-equilibrium [1]. Many-body perturbation theory provides a robust and systematic first-principles framework to develop formalisms describing these complicated multiparticle interactions and excited-state phenomena. Through decades of developments, the *ab initio* GW method [2,3] and the GW-Bethe-Salpeter-equation (GW-BSE) approach [4-7] have become the state of the art for accurately describing quasiparticle band structures, optical excitations, and excitonic effects in materials, mostly focusing on e - e and e - h interactions. Very recently, new many-body approaches have been developed to extend the descriptions to broader phenomena, including the e -ph coupling using GW perturbation theory (GWPT) [8-12], and nonequilibrium excited-state dynamics using the time-dependent GW (TDGW) approach [13-17]. These advanced theories, methods, and their high-performance implementations enable us to explore exciting frontier topics such as field-driven coherent phases, self-trapped excitons, excited-state dynamics at ultrafast timescale, and correlated e -ph interaction and nonequilibrium exciton-phonon coupling in quantum materials, two-dimensional (2D) materials and moiré superlattices.

This INCITE proposal centers around efforts in the DOE Computational Materials Sciences (CMS) Center for Computational Study of Excited-State Phenomena in Energy Materials (C2SEPEM), with the vision to develop new *ab initio* theories, methods, algorithms, and open-source software to elucidate and predict excited-state phenomena. The central software package is the BerkeleyGW code [18,19], which has recently demonstrated excellent scalability (for GW and GWPT) up to nearly the full scale of Frontier (OLCF) and Aurora (ALCF) – the two leading exascale high-performance computing (HPC) facilities – with excellent peak performance of over 500 PetaFLOP/s [20]. The combination of the demonstrated exascale capability and the advanced approaches for various multiparticle interactions and excited-state phenomena opens up many exciting opportunities in both addressing *fundamental scientific questions* and achieving *exascale computing for materials science*:

- This proposal focuses on four thrust areas, covering **emerging topics of phonon-coupled excited-state phenomena, correlated electronic and excitonic states in moiré superlattices, and nonequilibrium dynamics of excited states**. The role of self-energy and excitonic effects in these phenomena are elusive, and their understanding is key to engineering excited states of quantum materials. The GW, GW-BSE, GWPT, and TDGW methods are uniquely positioned to address these questions reliably from first principles because of their accuracy and systematicity at the many-electron level.
- **The GW, GW-BSE, GWPT, and TDGW methods are computationally demanding**, being a few more orders of magnitude costly than standard density functional theory (DFT) based methods. At the same time, these many-body methods are particularly well suited for exascale computing, as they exhibit a high degree of parallelism and intensive computational kernels that can be greatly accelerated on modern GPU architectures.

In this proposal, we aim to push forward the fundamental understanding of excited-state phenomena in quantum materials using advanced *ab initio* many-body computational approaches, fully utilizing the exascale capabilities of Frontier and Aurora. The expected achievements will redefine the boundaries of both the excited states research and the materials science computation at scale. Below, we elaborate in details: i) *expected advances*, ii) *state-of-the-art and competing work*, and iii) *previous INCITE awards*.

1.1. Expected Advances

We propose *four* strongly coherent Thrust Areas (**Fig. 1**). These Thrust Areas will advance the fundamental understanding of important questions on the roles of various correlated multiparticle

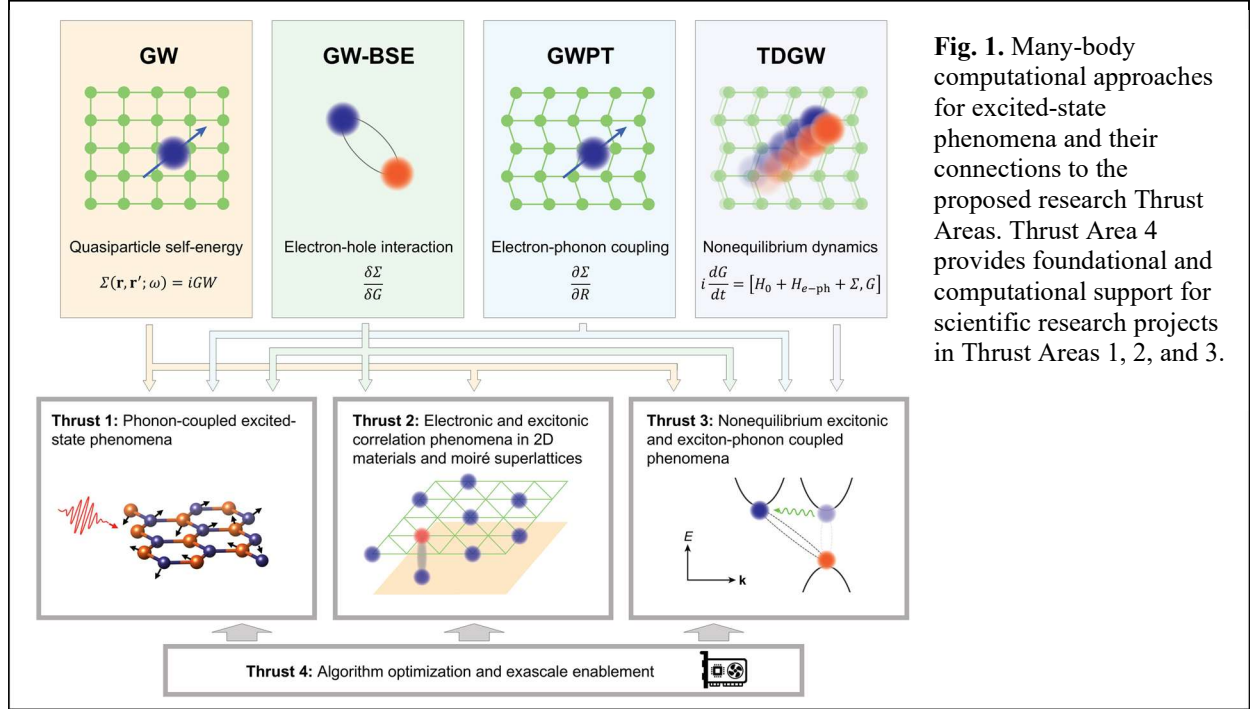


Fig. 1. Many-body computational approaches for excited-state phenomena and their connections to the proposed research Thrust Areas. Thrust Area 4 provides foundational and computational support for scientific research projects in Thrust Areas 1, 2, and 3.

interactions and excitations in emergent quantum phenomena, develop new correlated and nonequilibrium computational approaches and infrastructures, and extend the exascale capability of many-body quantum simulations to a broader range of methodologies.

Thrust Area 1: Phonon-coupled excited-state phenomena. This thrust area aims to address a series of phenomena where the coupling between the ionic degrees of freedom and electrons or excitons is essential:

- *Self-energy (many-electron) effects in e-ph coupling phenomena* are recognized to be critical in phonon-mediated superconductivity, transport, and polaron formation. Using our newly developed GWPT method (going beyond the standard density-functional perturbation theory (DFPT)), we will identify the role of the many-electron self-energy effects in the long-range Fröhlich *e-ph* interaction.
- *Excited-state forces and vibrational properties* are essential to light-induced structural responses such as self-trapped excitons and photocatalytic chemical reaction pathways. Excited-state forces, and more generally the vibrational properties (including phonons), can be accurately determined from the GW-BSE gradients to capture the responses to ionic displacements.
- *Exciton-phonon coupling and phonon-screened excitonic responses* play an important role in the optical response and energy-conversion efficiency, especially at finite temperatures. By extending our recent development of phonon screening in excitons, this proposal will further incorporate polaronic self-energies to provide a unified *ab initio* picture of *e-ph* effects in optical excitations.

Thrust Area 2: Electronic and excitonic correlation phenomena in 2D materials and moiré superlattices. This thrust area aims to address emergent correlated phenomena including the (generalized) Wigner crystals, Wigner crystalline excitons (WCEs), and excitonic insulators in 2D and moiré superlattice systems:

- *WCEs are moiré excitons arising from generalized Wigner crystals*, and they demonstrate strong correlation effects in the excited states and dictate the exciton dynamics and condensations. We have recently enabled direct GW-BSE calculations for system up to $\sim 10,000$ atoms to capture WCEs. This research direction will reveal and engineer the interplay between the strong *e-h* correlation and the moiré interaction landscape.
- *Excitonic insulators and exciton condensation* are exotic coherent phases driven by *e-h* interactions. Our recent efforts have successfully established an *ab initio* methodology based on GW-BSE and a

Bardeen-Cooper-Schrieffer (BCS)-type many-body formalism. We propose to identify potential candidates of excitonic insulators and to investigate the consequences of exciton condensations, such as charge-density waves.

Thrust Area 3: Nonequilibrium excitonic and exciton-phonon coupled phenomena. This thrust area aims to develop and apply TDGW approaches incorporating phonon dynamics, which enable nonequilibrium descriptions of exciton-phonon coupled processes:

- *Nonequilibrium exciton-phonon coupling* drives the dynamical scattering and decays of excitons via phonons. Through our newly developed time-dependent adiabatic GW (TD-aGW) method with phonons included (TD-aGW-ph), we will investigate how optically excited zero-momentum excitons scattered into finite-momentum ones in 2D materials. Moreover, higher-order phonon coupling effects will be captured through a frozen-phonon TD-aGW scheme. We will also apply a singular value decomposition (SVD) data-compression approach to accelerate TDGW calculations.
- *Dynamical formation processes of polarons and excitonic polarons* are key to the energy transfer and trapping mechanisms in materials. Our TD-aGW-ph method naturally describes such dynamical processes. We will investigate the nonequilibrium formation processes of polarons and excitonic polarons (self-trapped excitons) in polar semiconductors and halide perovskites, paving ways for optical engineering excitonic states.

Thrust Area 4: Algorithm optimization and exascale enablement. This thrust area aims to continue the exascale enablement of all core components of BerkeleyGW:

- *Exascale enablement* has been excellently demonstrated in the GW and GWPT functionalities, with up to full Frontier and 87.5% of Aurora, reaching over 500 PetaFLOP/s of peak performance. We propose to enable a similar level of exascale capability for the core computing modules of the GW-BSE and TDGW approaches.

1.2. State-of-the-Art and Competing Work

The research of our team focuses on excited-state phenomena including quasiparticle excitations, optical properties, time-dependent and nonequilibrium phenomena, multiparticle excitations, and coupling between phonons and correlated electrons and excitons. In recent years, approaches such as GW and GW-BSE methods, and software based on many-body perturbation theory, have become widely developed and well accepted as standard tools for materials research worldwide, with popular software packages including BerkeleyGW [18], WEST [21], Yambo [22], ABINIT [23], and VASP [24] codes. Moreover, besides BerkeleyGW, competitive progress on nonequilibrium many-body approaches has been made with efforts within the Exciting [25], CHEERS [26], and FHI-aims [27], and Hefei-NAMD [29] codes. For the e -ph coupling studies, prevailing and successful *ab initio* calculations are often carried out through the combination of DFPT (implemented in, e.g., Quantum ESPRESSO [29] and ABINIT [23]) and Wannier interpolation (implemented in, e.g., EPW [30], PERTURBO [31], Phoebe [32], and EPIq [33]).

Many of the existing implementations focus on particular aspects of the excited-state phenomena (e.g., either optical phenomena or e -ph coupling) or contain inconsistencies in level of theories (e.g., the exciton-phonon coupling is commonly computed with a GW-level e - h interaction but DFT-level e -ph matrix elements). Moreover, the computation of many-body interactions in real materials systems requires efficient and scalable code implementations capable to fully utilize large-scale HPC resources.

Originality in scientific insights and computational capabilities: Our expertise and the exascale-ready capability of BerkeleyGW excellently address these challenges through **two unique aspects**:

- **Comprehensive coverage of core interactions and dynamical processes at the many-body level.**

Within BerkeleyGW, the e - e , e - h , and e -ph interactions can be systematically computed with GW, GW-BSE, and GWPT methods, all *at a consistent many-body level*. Combining with these interaction ingredients, the nonequilibrium processes can be simulated with TD-aGW and TD-aGW-ph approaches, naturally enabling studies of excitonic and exciton-phonon coupled dynamics. Many of these cutting-edge methods are originated by team members of this proposal.

As shown in **Fig. 1**, the core computing components of GW, GW-BSE, GWPT, and TDGW are coherently interconnected to support the Thrust Areas. Our proposed research will reveal **deep scientific insights** in a broad range of excited-state phenomena both in- and out-of-equilibrium, including excited-state forces and phonon-coupled dynamics, formation of polarons and excitonic polarons, strongly correlated physics in WCEs and coherent exciton condensations, and field-driven optical phenomena.

- **Excellent scalability and performance of BerkeleyGW on exascale HPC systems.**

BerkeleyGW has been a leading excited-state software package among existing codes. Through previous INCITE Awards, our efforts have led to GPU-accelerated implementations of the GW, GW-BSE, and GWPT functionalities on Summit (NVIDIA GPU architecture), Frontier (AMD GPU architecture), and Aurora (Intel GPU architecture) with excellent portability across all major GPU hardware. The BerkeleyGW implementation and performance for Summit was selected as a Finalist for 2020 ACM Gordon Bell Prize [19]. Very recently, we have achieved over 500 PetaFLOP/s peak performance on both exascale HPC systems Frontier and Aurora, reaching ~30–40% of theoretical peak in the core computing kernel for GW and GWPT [20]. This proposal will further enable exascale capabilities for other key computing components including GW-BSE and TDGW, promising a new state of the art in scalability and new physics to be uncovered with this unique set of approaches.

1.3. Previous INCITE Awards

- **Li** and **Del Ben** have been awarded, as PI and Co-PI, an ongoing INCITE Award (2024 – 2025) titled *Exascale Simulation of Correlated Electron-Phonon Coupling in Quantum Materials*.
- **Del Ben** and **Qiu** have participated in a previous INCITE Award (2023) as Co-PIs. This INCITE Award was titled *Photophysics of Excitons in Low Dimensional Organic-Inorganic Semiconductors*, led by Marina Filip (Oxford).
- **Li**, **Del Ben**, **Jornada**, **Louie**, **Neaton**, and **Qiu** have participated a previous INCITE Award (2020 – 2021). This INCITE Award was titled *Novel Methods for Complex Excited-State Phenomena in Functional Materials*, led by Jack R. Deslippe (LBNL).

The previous INCITE Awards have enabled the development of the GPU-accelerated BerkeleyGW code for the GW, GW-BSE, and GWPT methods. Its successful scalability to full Summit (OLCF) with 105.9 PetaFLOP/s performance was recognized as a Finalist for 2020 Gordon Bell Prize [19]. Moreover, through the current INCITE Award, we have achieved true portability of BerkeleyGW across all major GPU hardware architectures, namely NVIDIA, AMD, and Intel, using both open programming models (OpenACC, OpenMP) and hardware-specific programming models (CUDA, HIP, SYCL). The new development has successfully scaled the GW and GWPT calculations to full Frontier (9,408 nodes, 75,264 GPUs) and 87.5% of Aurora (9,296 nodes, 111,552 GPUs), reaching over 500 PetaFLOP/s on both exascale supercomputers [20]. The massive computational resources offered by the INCITE Awards and the highly scalable implementations of various many-body approaches have led to exciting scientific discoveries and understanding. Through previous INCITE Awards, the team members in this proposal have produced 17 publications [9,11,19,20,34-46], including 2 *Phys. Rev. Lett.*, 1 *Nature*, 2 *Nat. Commun.*, 1 *Nano Lett.*, 1 *ACS Nano*, and 1 *Supercomputing Conference (SC) proceedings* (2020 ACM Gordon Bell Finalist).

This proposal serves as a significant expansion building on the prior successes achieved for exascale many-body excited-state calculations. We plan to extend the application cases of these advanced methods to study correlated and coherent excited-state phenomena, where complicated multiparticle interactions dominate the emergence of exotic phases and determine the functionalities of quantum materials. We will leverage the achieved exascale enablement in GW and GWPT methods to further scale the GW-BSE and TDGW methods to full machine at Frontier and Aurora. The new scientific research, method developments, and software developments proposed here will continue to place BerkeleyGW among the *best performant* materials science codes [19,20], serving as a powerful computational infrastructure to advance the scientific boundary for excited-state, nonequilibrium dynamics, and quantum materials research, benefiting the broad community.

2. RESEARCH OBJECTIVES AND MILESTONES

Thrust Area 1: Phonon-coupled excited-state phenomena (400,000 Node Hours)

2.1. Self-energy effects in long-range Fröhlich e -ph interaction (150,000 Node Hours)

Our recent development of GWPT [9,12] (Li, Jornada, Louie) has pushed forward *ab initio* e -ph computation by incorporating many-electron self-energy effects beyond DFPT, allowing for the calculation of e -ph coupling at unprecedented accuracy. GWPT computes the e -ph matrix elements at the GW level through direct construction of the first-order linear-response change in the self-energy operator, $\partial_{\mathbf{q}\nu}\Sigma$, where $\Sigma = iGW$ is in the GW approximation [2,3] (G : Green's function; W : screened Coulomb interaction). We have revealed critical roles of e -ph coupling in bismuthate [8], cuprate [9], nickelate [10], and kagome superconductors [11] on photoemission kinks (band renormalization) and superconductivity.

Besides metallic systems, e -ph coupling is critical in functional properties of semiconductors [47] such as transport and phonon-assisted optical absorption, as well as central to the formation of polarons and excitonic polarons. In polar semiconductors and insulators, the e -ph matrix elements have a divergence at small $|\mathbf{q}|$ (i.e., large spatial distance) of the form $\sim 1/|\mathbf{q}|$ for the longitudinal optical (LO) phonon modes (as illustrated in the Fröhlich interaction), which needs to be appropriately treated. Standard approaches [48-51] interpolate the short-range e -ph coupling (g^S) using Wannier functions, and model the long-range coupling (g^L) by dipole-induced classical Coulomb potential. However, it is **unclear** whether this approach applies at the GWPT level and what role does self-energy effects play in Fröhlich interaction.

Goals: We (Li, Louie, Del Ben) aim to address the fundamental role of self-energy effects in the e -ph coupling, focusing on polar semiconductors. We propose to develop appropriate methods to enable Wannier interpolation of the GWPT Fröhlich-type e -ph matrix elements, which essentially completes the GWPT infrastructure of e -ph coupling computation for general materials.

Theoretical and computational methods: This project mainly involves the GW and GWPT calculations implemented in BerkeleyGW and the Wannier interpolation scheme implemented in the EPW code with our in-house modifications. Li has performed preliminary DFPT and GWPT studies of SiC, which show different long-range limits of the e -ph matrix elements for the LO modes (Fig. 2). We have preliminarily derived and implemented an analytical expression of the GTPW long-range e -ph vertex. Fig. 2 shows that the interpolation scheme well captures the long-range diverging behaviors of the GWPT matrix elements. In this proposal, we will develop both the dipolar and quadrupolar GWPT e -ph coupling interpolations. This project sets up a foundation for the polaron-related projects (see Sec. 2.2, 2.3, 2.7) for a full many-body GW description.

Milestones: In Year 1, we will complete the implementation of the analytical form of the self-energy term in dipolar long-range e -ph interaction and apply it to study various materials of interests including GaN, boron nitride, and halide perovskites. In Year 2, we will further develop the treatments for quadrupolar interactions and different asymptotic behaviors in lower dimensions.

2.2. Excited-state forces and vibrational properties (150,000 Node Hours)

Optical excitations can induce significant atomic motions in a variety of semiconductors and molecules. Understanding these phenomena can unlock a new degree of tunability in photocatalysis, optical switches, and quantum sensors [52-54]. To obtain excited forces and vibrational properties such as phonon spectrum, the first- and second-order derivatives of the GW-BSE e - h interaction kernel and exciton excitation energy are needed. Excited-state forces and derivatives can be analytically obtained [55,56] based

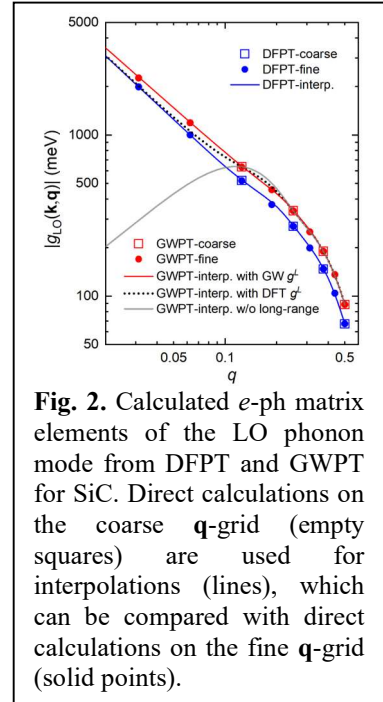
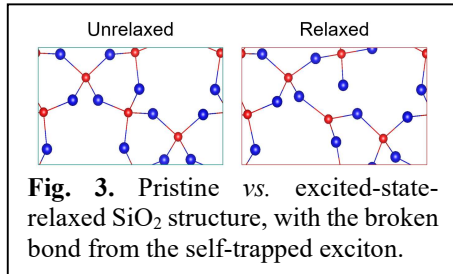


Fig. 2. Calculated e -ph matrix elements of the LO phonon mode from DFPT and GWPT for SiC. Direct calculations on the coarse $\mathbf{q}$ -grid (empty squares) are used for interpolations (lines), which can be compared with direct calculations on the fine $\mathbf{q}$ -grid (solid points).

on GW-BSE, DFPT, and GWPT calculations, with a constant screening approximation imposed. Alternatively, finite-difference (frozen-phonon) approach offers controllable numerical accuracy towards exact derivatives. Our team has preliminarily developed and implemented both strategies.



doped $\text{Cs}_2\text{AgInCl}_6$ [58,59], where self-trapped excitons are suggested to be important. Through excited-state forces, we propose to perform structural relaxations, emission Stokes shifts, and to identify dominant phonon modes drive the self-trapping processes.

Goals: We (Jornada, Qiu) aim to comprehensively develop the computational infrastructure for excited-state forces and vibrational properties based on GW-BSE method to obtain gradient quantities. We will apply these approaches to systematically study excited-state vibrational properties at the many-body level, focusing on single-photon emitter in 2D defects, photocatalyst, and self-trapped excitons.

Theoretical and computational methods: Jornada has recently developed a finite-difference (frozen-phonon) approach that numerically evaluates the first and second derivatives of expectation values of operators evaluated for excited states obtained from GW-BSE calculations. We have benchmarked our approach on a variety of small systems (see Fig. 3 for SiO_2) and shown excellent agreement to both previous calculations using other methods and experiments [60]. Moreover, Qiu and Li have preliminarily implemented the linear-response analytical form of excited-state force, where constant screening approximation is assumed. The two approaches are complementary to each other.

Milestones: In Year 1, we will benchmark and cross-validate the analytical and numerical derivative approaches for excited-state forces, and focus on the excited-state structural relaxation calculations. In Year 2, we will compute and analyze the concrete phenomena in the proposed materials systems.

2.3. Exciton-phonon coupling and phonon-screened excitonic responses (100,000 Node Hours)

Exciton-phonon coupling plays an important role in the optical response and energy-conversion processes in a broad range of quantum materials. Dynamical screening effects from phonons can potentially reshape the optical and excitonic properties, but are often neglected in prior studies with standard GW-BSE approaches. Our *ab initio* work (Neaton, Louie, Li) [61-63] has successfully incorporated phonon screening kernel into the GW-BSE formalism, revealing a renormalization in exciton binding energies by tens of percent and enabling phonon absorption- and emission-driven dissociation channels of excitons into hot carriers. Fig. 4 shows the impact of phonon screening on the lowest exciton in BiVO_4 , with significant renormalization of the binding energy and the ultrafast spontaneous dissociation.

The importance of phonon screening motivates further exciton-phonon method development, and also offers numerous opportunities for application to complex materials. Particularly, we propose to further extend our exciton-phonon GW-BSE formalism to include polaronic self-energies into the e - h kernel, providing a unified *ab initio* picture of the effect of e -ph coupling in optical excitations. We plan to investigate materials for which the discrepancy between theoretical calculations and experimental measurements of the exciton binding energy are large and where polaronic effects are non-negligible. Due to the dense sampling required for phonon ($\mathbf{q}$) and exciton ($\mathbf{Q}$) wavevectors, we will leverage our recently developed maximally localized exciton Wannier functions (MLXWFs) [64] (Neaton, Jornada, Qiu) to efficiently interpolate and sample the exciton-phonon coupling matrix elements and to facilitate interpretations of the results using the compact real-space basis functions.

Goals: We (Neaton) aim to further enable advanced features for phonon-screening e - h kernel, including spin-orbit coupling, nonuniform Brillouin-zone sampling, and polaronic contributions. We plan

to identify phonon-screening effects in the optical responses of lead halide perovskites, and clarify role of polarons in reshaping the exciton-phonon coupling in polar systems including MgO, SrTiO₃, and BiVO₄.

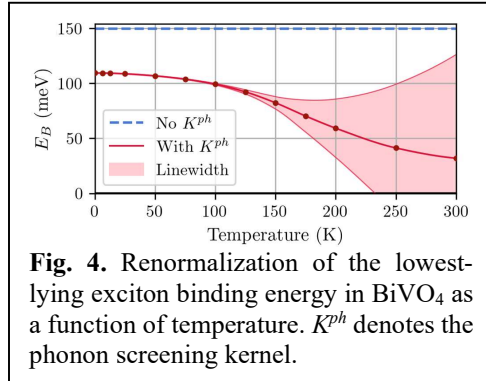


Fig. 4. Renormalization of the lowest-lying exciton binding energy in BiVO₄ as a function of temperature. K^{ph} denotes the phonon screening kernel.

Theoretical and computational methods: The dynamical screening effects from phonon vibrations can be formulated perturbatively with phonon propagators and e -ph matrix elements. Finite- $\mathbf{Q}$ GW-BSE solutions [65] are needed, accompanied by DFPT and GWPT calculations on the same $\mathbf{q}$ mesh. The direct first-principles calculations will be performed on a coarse grid and then interpolated onto a fine grid using electron and exciton Wannier functions. Then, for each system, the phonon screening kernel will be constructed for each temperature. Because dynamical effects in phonon screening in the BSE are not negligible, the final resulting GW-BSE solutions must be obtained self-consistently. The massive

parallelization nature in the exciton $\mathbf{Q}$ and phonon $\mathbf{q}$ grids suits excellently the exascale supercomputers.

Milestones: In Year 1, we will focus on enabling the advanced features in the phonon screening kernel and apply the established method for systematic investigation of a series of proposed materials. In Year 2, we will incorporate the polaronic effects in phonon screening and study its impact.

Thrust Area 2: Electronic and excitonic correlation phenomena in 2D materials and moiré superlattices (300,000 Node Hours)

2.4. Strong excitonic correlation effects in Wigner crystalline excitons in 2D moiré superlattices (180,000 Node Hours)

Moiré superlattices host strong Coulomb interactions residing in the narrow electron bands, leading to correlated insulating states at low doping, known as generalized Wigner crystals [66-68]. Upon optical excitations, the formation of moiré excitons is fundamentally shaped by the Wigner-crystal ground states, leading to exotic phases such as exciton insulators [69] and exciton density waves [70]. The Wigner crystalline excitons (WCEs) manifest intertwined e - e and e - h correlations in the ground and excited states.

Despite a growing collection of experimental signatures and phenomena, the interplay among Wigner crystallization, moiré superlattice potentials, and excitonic correlations remains elusive with limited microscopic understanding. A first-principles approach has been deemed intractable due to the complex nature of the mixed boson-fermion systems and their large sizes. We (Li, Del Ben, Louie) have successfully overcome these theoretical challenges [36,71] in our preliminary studies of MoSe₂/MoS₂ moiré superlattices. Here, we developed a technique to create Wigner-crystal ground states using a symmetry-broken DFT+ U method. Moreover, we enabled first-principles GW-BSE calculations of WCEs with $\sim 10,000$ atoms on Aurora using thousands of nodes. Our results [36] surprisingly uncover the propagation of correlation effects from ground states to excited states, shaping the characteristics of WCEs (Fig. 5).

Goals: We (Li, Del Ben, Louie) aim to systematically study the WCEs in various moiré superlattice systems of transition-metal dichalcogenides. We will construct the dipolar field distribution using the exciton solutions from GW-BSE. The dipolar interactions among excitons dictate the exciton condensation conditions and are key to understanding the complicated mixed boson-fermion correlations. Moreover, we (Louie) will investigate alternative DFT approaches beyond DFT+ U , such as using hybrid functionals, to create ground-state (generalized) Wigner crystals of electrons.

Theoretical and computational methods: To enable *direct* GW-BSE calculations for 2D systems with $\sim 10,000$ atoms, several computational bottlenecks have been successfully addressed [36] (Li, Del Ben, Louie). One of the major computational components in the GW-BSE calculation is the construction of the inverse dielectric matrix $\epsilon^{-1}(\mathbf{r}, \mathbf{r}')$, which is critical in describing the screening environment of the e - h interaction. Here, we separate the computation of dielectric matrix into low-energy and high-energy parts. The low-energy polarizability is directly constructed with moiré wavefunctions which contains doping-induced Wigner-crystal states, whereas the high-energy part is folded from unit-cell polarizability. This strategy significantly reduces the computational cost. Additionally, the computation of e - h BSE kernel is

highly memory demanding (distribution over data of a few TB). We redesigned the implementation by batching the large-size tensor and recompute necessary intermediates for each batch, with a low re-computation overhead through replication of data depending on available memory. The new implementation shows much improved time-to-solutions on thousands of Aurora nodes with low communication cost.

The GW calculations of the band structure are critical to understand the electronic correlations but computationally formidable as well. We (**Jornada**) have developed two complementary advancements that reduce the cost of the standard GW approach for large moiré systems. The first advancement is an algorithm that stochastically compresses the wavefunctions, termed stochastic pseudobands, which reduces the overall scaling of the GW calculation from $O(N^4)$ to between $O(N^2)$ and $O(N^3)$ [72]. The second advancement alleviates the difficulty of generating sufficiently many mean-field states to converge the GW calculation by making use of a diagonalization-free Chebyshev-Jackson expansion algorithm to directly construct the compressed stochastic pseudoband states in only $O(N^2)$ time. Together, they enable routine GW band-structure calculations for moiré systems involving thousands of atoms.

Milestones: In Year 1, we aim to apply the existing GW and GW-BSE implementations to study WCEs in selected moiré superlattices. In Year 2, we aim to achieve the exascale enablement of the GW-BSE calculations to full scale of Frontier of Aurora (currently at few-thousand node scale).

2.5. Excitonic insulators and exciton condensation (120,000 Node Hours)

The excitonic insulator phase is a coherent superfluid state [73]. Recent momentum-resolved electron-energy-loss spectroscopy, angle-resolved photoemission spectroscopy (ARPES), and scanning-tunneling microscopy studies have revealed that several small-overlap semimetals, e.g., monolayer 1T'-WTe₂, monolayer 1T-HfTe₂ and 1T-TiSe₂, enter a charge-density-wave (CDW) phase below a critical temperature [74–76]. Interestingly, the emergence of the CDW phase is conjectured to be tightly intertwined with the excitonic-insulator condensation phase, where e - h interactions are the driving force. However, whether the CDW is driven purely by excitons, purely by phonons, or by a cooperative exciton-phonon mechanism remains debated. Accurate descriptions of the excitonic interaction and its consequences on the formation and condensation of excitonic insulators is critical for both fundamental physics understanding and creating devices that can switch between metallic and insulating states.

Goals: We (**Louie**) aim to systematically study the possible excitonic insulator phases in monolayer 1T-HfTe₂ and 1T'-WTe₂, where the CDW phases correspond to the instability at finite momentum. We will

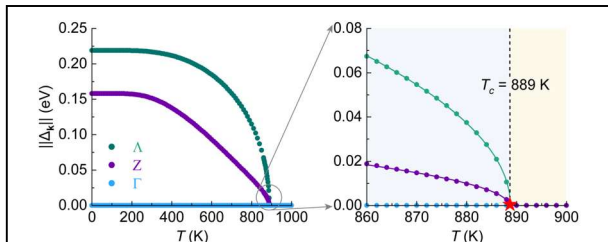


Fig. 6. Temperature dependence of order parameter at different $\mathbf{k}$ -points, predicting the critical temperature for the excitonic insulator in 1T'-MoSe₂.

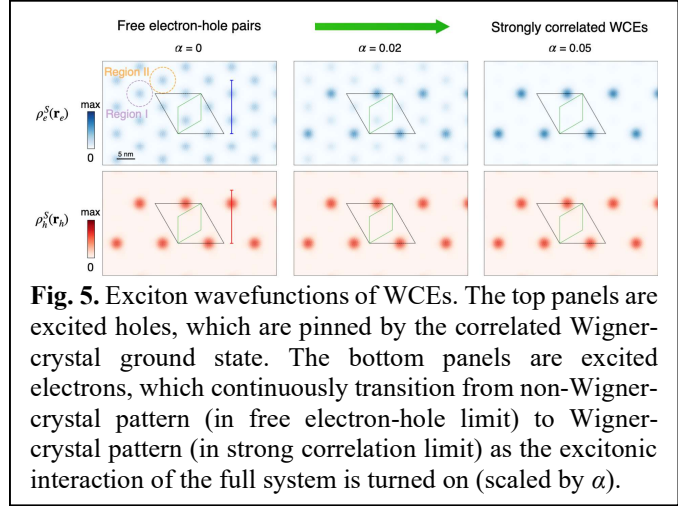


Fig. 5. Exciton wavefunctions of WCEs. The top panels are excited holes, which are pinned by the correlated Wigner-crystal ground state. The bottom panels are excited electrons, which continuously transition from non-Wigner-crystal pattern (in free electron-hole limit) to Wigner-crystal pattern (in strong correlation limit) as the excitonic interaction of the full system is turned on (scaled by α).

compute their excitonic insulator order parameter and predict its temperature dependence, and compare the calculated spectroscopic signatures with tunneling and photoemission experiments.

Theoretical and computational methods:

The excitonic insulator phase can be described following a BCS-like theory, where a many-body Hamiltonian includes the two-body e - h interaction kernel. We (**Louie**) have recently developed an approach that combines first-principles interaction matrix elements from GW-BSE and a many-body

Hamiltonian, and has predicted a spontaneous excitonic insulator phase in 1T'-MoSe₂ [77]. Here, we solve for the order parameter for the excitonic insulator phase, whose temperature dependence gives rise to the critical temperature for the phase transition between band insulator and excitonic insulator (see Fig. 6). In our first work, we focused on the subspace of e - h pairs with $\mathbf{Q} = 0$. To study the possibility of excitonic interaction driven CDW phases, especially those with enlarged supercells, finite- $\mathbf{Q}$ GW-BSE calculations and an extended many-body Hamiltonian with $\mathbf{Q}$ -dependence needs to be developed. We propose to generalize our method to capture full momentum ($\mathbf{Q}$) dependence in solving for the excitonic insulator phases. This project can delineate the mechanisms for excitonic insulator candidates, and address some ongoing debates.

Milestones: In Year 1, we will perform the GW-BSE calculations for candidate materials, and develop the finite- $\mathbf{Q}$ excitonic insulator many-body Hamiltonian solver. In Year 2, we will perform systematic calculations and analysis on the excitonic origin of the CDW phases.

Thrust Area 3: Nonequilibrium excitonic and exciton-phonon coupled phenomena (100,000 Node Hours)

2.6. Nonequilibrium exciton-phonon coupling in 2D materials (50,000 Node Hours)

In the last two decades, ultrafast laser techniques have seen considerable developments, and consequently, ultrafast spectroscopic techniques, including pump-probe transient absorption, solid-state high-harmonic generation, and time-resolved ARPES (tr-ARPES) have greatly broadened the exploration of novel non-equilibrium physics in condensed matter systems. One emerging theoretical approach is time-dependent adiabatic GW (TD-aGW) [13,16] based on the nonequilibrium Green's function framework [13-16]. However, the currently established TD-aGW formalism does not incorporate effects from phonons, which serve as important scattering channels and play critical roles in coherent and incoherent coupling to electrons, excitons, and general excitations. We (Li, Louie, Del Ben, Jornada) have recently developed a new method to include the coherent e -ph coupling into the TD-aGW formalism and implementation, which allows for **finite-momentum dynamics involving both e - h and e -ph interactions**. This new method is denoted as TD-aGW-ph.

Recent pump-probe tr-ARPES experiments on MoS₂ have resolved clearly that zero-momentum excitons can be scattered to finite-momentum excitons at a timescale of ~ 1 ps [78], involving phonon-coupled dynamics. The TD-aGW-ph approach provides an excellent opportunity to gain deep insights in these nonequilibrium phenomena and to design new optoelectronic devices or heterostructures. This project aims to use TD-aGW-ph to understand how carriers and excitons are excited and scattered by phonons in a

short time scale ($\sim$ ps) in real time.

Goals: We (Li, Louie, Del Ben, Jornada) aim to apply the TD-aGW-ph method to study the dynamical exciton-phonon coupling process in monolayer MoS₂ by directly simulating the time evolution of exciton states after light pump. We will extract the lifetime information and the scattering pathways in the momentum space.

Theoretical and computational methods: The TD-aGW-ph formalism includes e -ph coupling and e - h interactions coherently across the full Brillouin zone. The computational cost is quite heavy as there are typically $O(1,000)$ phonon wavevectors involved, which lead to the effective cost as thousands of coupled TD-aGW equations of motion evolving simultaneously. We plan to address the computational bottleneck via two ways. First, we (Li, Del Ben, Louie) aim at its HPC optimization and exascale adaptation. The heavy communication can be overlapped with the computation modules. Second, we propose to use our recently developed data compression technique based on SVD [79] (Qiu) to reduce the problem dimensions. The compressed BGW e - h kernel can significantly speed up the TD-aGW calculations by orders of magnitude.

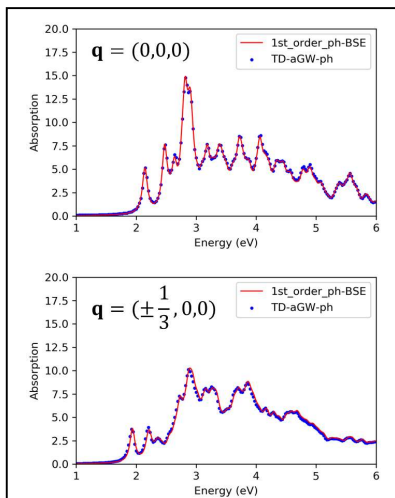


Fig. 7. Verification of TD-aGW-ph with frozen-phonon-based BSE calculations, with phonon effects limited to the first order.

The TD-aGW-ph formalism is derived based on the linear-response theory, such that all calculations can be performed within a single primitive unit cell. However, higher-order exciton-phonon coupling can be important, especially at elevated temperatures. To complement TD-aGW-ph method and to investigate nonlinear exciton-phonon coupling, we (**Louie**) propose to combine the special-displacement thermal sampling [80,81], which generates a single representative structural configuration in a supercell at a given temperature, with the GW-BSE and TD-aGW calculations. The frozen-phonon TD-aGW and TD-aGW-ph calculations can validate each other as well, as shown in **Fig. 7** with an example of monolayer MoS₂.

Milestones: In Year 1, we will apply the TD-aGW-ph and frozen-phonon TD-aGW method to study the dynamical exciton-phonon scattering in 2D materials. In Year 2, we will combine SVD data compression to accelerate the nonequilibrium simulations.

2.7. Dynamical formation processes of polarons and excitonic polarons (50,000 Node Hours)

Polarons are quasiparticles arising from e -ph and exciton-phonon interaction. Upon carrier doping or optical stimuli, polarons or excitonic polarons can arise with locally trapped charges or excitons by the structural distortions. A deep understanding in their dynamical formation processes offers opportunities to achieve external control of materials' structural, transport, and optical properties via gating or light modulation. Our recently developed TD-aGW-ph offers an exciting opportunity to directly simulate the dynamical formation processes of polarons [82] and excitonic polarons [83,84]. All essential microscopic ingredients are naturally included in the equation of motion. Moreover, with our developed interpolation scheme of Fröhlich interaction at the GWPT level (see **Sec. 2.1**), the theoretical framework is consistently at the many-electron GW level.

Goals: We (**Li, Del Ben, Louie**) aim to develop the polaron and excitonic polaron functionalities based on TD-aGW-ph. We will apply the new approach to study the dynamical formation processes in polar materials such as LiF and MgO, as well as halide perovskites, where polarons are critical to their optical absorption processes.

Theoretical and computational methods: The TD-aGW-ph formalism currently assumes a phonon bath where the solution of the phonon equation of motion is a harmonic oscillator, and the phonon population is solely determined by the temperature. To enable simulations of polarons and excitonic polarons, electron feedback to the phonon equation of motions needs to be incorporated. This can be achieved via adding an e -ph coupling term in the phonon equation of motion, such that the phonon population is modulated by the e -ph and exciton-phonon coupling. Particularly, this formalism captures light-induced structure change effects in the form of polarons under realistic physical conditions.

Milestones: In Year 1, we will implement the e -ph coupling term in the phonon equation of motion and carry out verification and benchmark calculations. In Year 2, we will apply the developed implementation to study the dynamical formation processes of polarons and excitonic polarons.

Thrust Area 4: Algorithm optimization and exascale enablement (150,000 Node Hours)

2.8. Full exascale enablement of core components of BerkeleyGW (150,000 Node Hours)

Excited-state calculations such as GW, GW-BSE, GWPT, and TDGW, are highly computationally demanding. Compared with the prevailing approaches such as DFT, the many-body calculations often require 2 – 4 orders of magnitude more resources. Our proposed research projects are not possible without the significant support at the scale that INCITE offers. BerkeleyGW has been at the forefront of the HPC optimization and utilization for years. In 2020, BerkeleyGW achieved a performance 105.9 PetaFLOP/s with 53% of the theoretical peak running full scale on Summit, and the work (**Del Ben, Louie, Li, Jornada**) has been selected as a Finalist for 2020 ACM Gordon Bell Prize [19]. Very recently, we (**Del Ben, Li, Jornada, Louie**) have successfully scaled GW and GWPT calculations to the full Frontier and 87.5% of Aurora, achieving core kernel performance of over 500 PetaFLOP/s [20], *a 5-fold improvement in the span of 5 years*. Furthermore, we have achieved true performance portability across all major GPU architectures, including NVIDIA, AMD, and Intel GPUs, using both open programming models (OpenMP, OpenACC) and hardware-specific languages (CUDA for NVIDIA, HIP for AMD, SYCL for Intel). We aim to use this INCITE allocation to keep up with the BerkeleyGW performance optimization as new methods emerge.

Goals: We (**Del Ben, Li, Louie, Jornada, Qiu, Neaton**) aim to extend the exascale capability for advanced GW-BSE and TDGW components and their applications with BerkeleyGW.

Theoretical and computational methods: We have recently enabled GW-BSE calculations of $\sim 10,000$ atoms (**Sec. 2.4**) running with thousands of Aurora nodes. It is relatively straightforward to scale it up to near full machine scale and adapt the implementation to Frontier. The TDGW component is a CPU code at this moment. It currently implements a two-level parallelization scheme over both $\mathbf{k}$ and $\mathbf{Q}$, allowing for scaling up to tens of thousands of MPI tasks, placing TDGW at an ideal position for massive parallelization at exascale. We plan to first enable the GPU acceleration in the TDGW code via open programming models such as OpenMP and OpenACC, which fit excellently the many nested-loop code structure and offer flexible and fast development cycles. The main bottleneck is in the communication, as all equations of motions depend on others through the $\mathbf{Q}$ dependence. We will develop strategy to overlap communication with computation. Polaris (ALCF) will be used for initial developments and benchmarks. To further improve the performance, we will develop hardware-specific and optimized codes for particular GPU architectures, specifically for AMD and Intel GPUs on Frontier and Aurora, respectively.

Milestones: In Year 1, we will scale up the GW-BSE calculations to a large fraction to full Frontier and Aurora, and enable the GPU capability in the TDGW code. In Year 2, we aim to optimize the performance of TDGW via algorithmic designs and hardware-specific programming models.

3. TECHNICAL ASSESSMENT

The intensive computational workload for this INCITE proposal will use BerkeleyGW [18], a massively parallel open-source software package employed to study the excited state properties of electrons in materials by using GW, GW-BSE, GWPT, and TDGW methods. BerkeleyGW has shown the capability to scale up to tens of thousands of CPU nodes [85] and effectively utilizing GPU accelerators to up to over 100,000 GPU devices [20]. Prior INCITE Awards have provided critical supports for the exascale adaptation of BerkeleyGW. In 2020, BerkeleyGW has demonstrated strong scaling to the entire Summit (4,608 nodes or 27,648 NVIDIA GPUs), as well as achieving high fraction of peak performance of 52.7% at double-precision 105.9 PetaFLOP/s [19]. This work (**Del Ben, Louie, Li, Jornada**) was recognized as a Finalist for the ACM Gordon Bell Prize (2020). Moreover, very recently (**Del Ben, Li, Jornada, Louie**), BerkeleyGW has achieved high double-precision throughput of 558.3 PFLOP/s on Frontier (9,408 nodes or 75,264 AMD GPUs) and 500.97 PFLOP/s on Aurora (9,296 nodes or 111,552 Intel GPUs), corresponding to 31.04% and 39.39% of the theoretical and attainable peak, respectively [20]. Very large-scale of GW and GWPT calculations are demonstrated, with massive system applications including silicon (Si) divacancy (up to 2,742 atoms), boron nitride (BN) moiré bilayer with defect (867 atoms with vacuum), and lithium hydride (LiH) defect (up to 17,574 atoms). BerkeleyGW has a very broad user base across North America, South America, Europe, Asia, and Oceania. The code is heavily used on leadership-class HPC facilities.

BerkeleyGW has served as the main computational engine for the following programs:

- **INCITE Project:** *Exascale Simulation of Correlated Electron-Phonon Coupling in Quantum Materials*, PI Zhenglu Li (University of Southern California), 2024 – 2025.
- **INCITE Project:** *Photophysics of Excitons in Low Dimensional Organic-Inorganic Semiconductors*, PI Marina Filip (Oxford University), 2023.
- **INCITE Project:** *Novel Methods for Complex Excited-State Phenomena in Functional Materials*, PI Jack Deslippe (NERSC), 2020 – 2021.
- **ALCC Project:** *Predicting heterogeneous photocatalysts using large-scale ab initio calculations*, PI Felipe da Jornada (Stanford University), 2024 – 2025.
- **ALCC Project:** *Dynamics and decoherence of excited states in 2D systems for quantum technology*, PI Mauro Del Ben (LBNL), 2022-23.
- **MatML ESP Project:** *Many-Body Perturbation Theory Meets Machine Learning to Discover Singlet Fission Materials*, PI Noa Marom (Carnegie Mellon University), 2024.

The workflow of GW, GW-BSE, GWPT, and TDGW calculations implemented in BerkeleyGW employs a sequence of steps performed by interdependent stand-alone executables. Each executable has its own computational expenses, system size scaling, memory requirements, I/O requirements, data layout and parallelization strategies. These executables are named parabands, epsilon, sigma, kernel, absorption, and dynamics, listed by the order of execution. All GW, GW-BSE, and GWPT calculations run with BerkeleyGW will take advantage of the GPU capabilities on Frontier (OLCF) and Aurora (ALCF). Input and output requirements for BerkeleyGW are handled via the high-performance parallel I/O HDF5 library.

3.1. Use of Resources Required

Computational cost: We estimate costs for first-year milestones. Estimations are made according to the system size scaling of the algorithms and our previous calculations on Frontier and Aurora. The GW and GWPT calculations are highly scalable with GPUs up to the full machine scale of Frontier and Aurora, so typically we will request several thousands of nodes, running for a few hours, for most of the tasks. The GW-BSE calculations can already scale up to a few thousands of nodes and Aurora, which typically complete in a few hours for system sizes of $\sim 10,000$ atoms. For each project, the anticipated burning rate will follow a pattern of periods of peak usage. Noticing the multi-group nature of the team, we expect the overall burning rate to be relatively stable. The costs for the second-year milestones are estimated similarly.

3.1.1. Self-energy effects in long-range Fröhlich e -ph interaction (150,000 Node Hours). The most computation intensive part is GWPT, which typically costs two to three orders of magnitude more than DFPT, and is equivalent to the workload for $\sim O(1,000)$ GW calculations. We have performed a full GWPT calculation for SiC (2-atom cell) with $8 \times 8 \times 8$ $\mathbf{k}$ -grid and $\mathbf{q}$ -grid, which costs $\sim 10,000$ GPU Node Hours on Frontier. We aim to study several long-range polar materials in both bulk and 2D systems, such as boron nitride, GaN, SrTiO₃, and TiO₂. GWPT (single phonon mode) scales as $O(N_{atom}^4)$ in the system size (number of atoms), therefore there is a naive factor up to 40 in the cost, for, e.g., SrTiO₃. However, larger-cell systems typically require less $\mathbf{k}$ - and $\mathbf{q}$ -points sampling, and the overall cost scales as $O(N_k^2 N_q)$, therefore a sampling with $6 \times 6 \times 6$ $\mathbf{k}$ -grid and $\mathbf{q}$ -grid sampling for SrTiO₃ acquires an additional factor of $\left(\frac{6^3}{8^3}\right)^3 = 0.075$. Therefore, for SrTiO₃, the estimated cost is around 3 times of SiC, of $\sim 30,000$ Node Hours on Frontier and Aurora. We plan to study 3 – 5 materials with the size of 2 – 8 atoms. Therefore, we request a total of 150,000 Node Hours.

3.1.2. Excited-state forces and vibrational properties (150,000 Node Hours). The finite-difference GW-BSE calculations are computationally intensive, and our focus systems are with defects or molecule-solid interface, which are typically to the order of hundreds of atoms. The structural relaxation is an iterative process based on excited-state force computed at each iteration. The GW-BSE calculations at one iteration costs around 1,000 Node Hours. Typically, around ~ 20 iterations are needed for convergence. Hence one material system will cost around 20,000 Node Hours. We propose to study 3 types of defect system, and 5 elementary reaction steps (structural configuration dependent), and thus request 150,000 Node Hours.

3.1.3. Exciton-phonon coupling and phonon-screened excitonic responses (100,000 Node Hours). The finite- $\mathbf{Q}$ GW-BSE calculations are quite demanding due to hundreds of $\mathbf{Q}$ -points needed for finite-momentum excitons. For a small system MgO, the cost is 20,000 Node Hours. We proposal to study a series of materials including SrTiO₃, and BiVO₄ with bigger system sizes. Hence, we request a total of 100,000 Node Hours.

3.1.4. Strong excitonic correlation effects in Wigner crystalline excitons in 2D moiré superlattices (180,000 Node Hours). Based on our preliminary calculations on MoSe₂/MoS₂ moiré superlattice with $\sim 10,000$ atoms, the epsilon calculation costs around 12,000 Node Hours on Frontier, and the kernel calculation costs around 6,000 Node Hours, which are the heaviest workload. For each system, we aim to study three doping concentrations, bringing the cost to 60,000 Node Hours for one material system. We propose to study three different moiré superlattices, and thus request 180,000 Node Hours.

3.1.5. Excitonic insulators and exciton condensation (120,000 Node Hours). The excitonic insulator candidates require a very fine sampling of the Brillouin zone and finite- $\mathbf{Q}$ BSE calculations. We estimate

20,000 Node Hours for the GW calculations and 40,000 Node Hours for the BSE kernel calculations. Therefore, two systems request 120,000 Node Hours.

3.1.6. Nonequilibrium exciton-phonon coupling in 2D materials (50,000 Node Hours). The TD-aGW-ph calculations are quite demanding due to the full phonon scattering channels across the Brillouin zone. We estimate the cost based on our experience on CPU architecture. For our test with a coarse sampling with $12 \times 12 \times 1$ $\mathbf{Q}$ -points, the cost is around 5,000 CPU Node Hours. A realistic sampling requires at least $24 \times 24 \times 1$ sampling with a cubic scaling, leading to around 320,000 CPU Node Hours. Based on a reasonable acceleration of 20 times using GPU compared with CPU, the estimated cost is around 16,000 GPU Node Hours. We target around 2 – 3 systems together with costs for the development stage, and therefore request 50,000 Node Hours.

3.1.7. Dynamical formation processes of polarons and excitonic polarons (50,000 Node Hours). Using TD-aGW-ph, similar to the project above on the system sizes and computational scale, we request 50,000 Node Hours.

3.1.8. Full exascale enablement of core components of BerkeleyGW (150,000 Node Hours). We have demonstrated that our advanced algorithms can achieve a low time to solution around ~10–30 minutes, leading to a full-machine job costing around 5,000 Node Hours. Here, we estimate the resources for method development to be ~90,000 Node Hours and exascale scaling calculations to be ~ 60,000 Node Hours to collect weak and strong scaling data. We thus request a total of 150,000 Node Hours.

Data storage: We estimate the anticipated cumulative size of the stored data will be 120 – 150 TB in scratch and around 50 TB in the long-term archival storage. There are two categories of the data: i) wavefunctions and ii) critical output data including dielectric matrices, kernel matrices, and e -ph matrices. Wavefunctions are typically very large, e.g., a single wavefunction for a system with thousands of atoms can take up few TBs space. However, wavefunctions are temporary input data and can be regenerated with reasonable cost, and we mostly keep them in scratch with a lifetime of 2–3 months. The output data are typically in HDF5 format, and we save them in the long-term storage for analysis, with a lifetime of ~6 months. Data that will be transferred into Frontier and Aurora are mostly input wavefunctions (HDF5) generated by DFT codes on other computer resources. We transfer the BerkeleyGW output data (HDF5) out of Frontier and Aurora. We will mostly use Globus transfer infrastructure, which is efficient and convenient according to our recent experiences.

3.2. Computational Approach

BerkeleyGW implements GW, GW-BSE, GWPT, and TDGW many-body methods to study the excited state properties of materials. As the state-of-the-art, these first-principles calculations for complex materials applications are often perceived as prohibitive due to the expensive computational cost compared to the widely used and less accurate DFT approaches. For example, the standard GW method scales as $O(N^4)$ as compared to $O(N^2)$ – $O(N^3)$ of conventional DFT. The massive parallelization of BerkeleyGW, its efficient GPU utilization across all major GPU architectures (NVIDIA, AMD, Intel), and its capability to achieve high fraction of peak performance, allow for excited-state studies of a large variety of materials with excellent time to solution (tens of minutes for systems of few thousands of atoms).

The implementation in BerkeleyGW is mainly based on the deterministic approach, involving summation over many empty states. Despite being $O(N^4)$, the advantage of deterministic approach is its general applicability and capability of providing all levels of materials' information (e.g., dispersion relations, spectral functions, quasiparticle lifetimes, etc.). Additionally, from a computational point of view, the practical implementation of the deterministic approach offers several levels of parallelism, and the compact underlying computational kernels can greatly take advantage of GPU accelerations.

BerkeleyGW consists of several major modules that are exascale-ready or nearly ready, for carrying out the GW and GWPT (epsilon, sigma) and GW-BSE (kernel and absorption) calculations, along with the parabands module for the generation of many empty bands, and the dynamics module (in the fast development stage) for TDGW simulations. The HDF5 data format and associated libraries are used to

streamline the I/O requirements over the entire workflow. The following parallelization, programming models, and libraries are employed:

- Programming language: Fortran (main), C, C++, Python, Perl, Bash
- Hybrid parallelization and portable programming models:
 - ✓ Inter-node level: MPI
 - ✓ Intra-node level (many-core CPU): OpenMP
 - ✓ Accelerators (GPU, mainline): OpenMP-target, OpenACC
 - ✓ Accelerators (CPU, dedicated branches): CUDA (NVIDIA), HIP (AMD), SYCL (Intel)
- Libraries:
 - ✓ Required: BLAS, LAPACK, FFTW
 - ✓ Optional (recommended): ScaLAPACK, HDF5, ELPA, cuFFT, cuBLAS, ROCm, hipFFT, hipBLAS, OneAPI, PRIMME

All core modules for GW, GW-BSE, and GWPT calculations within BerkeleyGW have been ported to GPU, achieving excellent performance on various architectures (NVIDIA, AMD and Intel GPUs). We have implemented OpenMP-target and OpenACC programming models to ensure performance portability. In BerkeleyGW, epsilon computes the inverse dielectric matrix, and sigma computes the self-energy operator and its phonon-perturbed changes to obtain the GW quasiparticle energies and the GWPT e -ph matrix elements. For the GW-BSE calculations, the core computing modules are kernel and absorption, which have both been GPU-enabled. The kernel module computes e - h interaction matrix, and the absorption module diagonalizes exciton BSE Hamiltonian matrix.

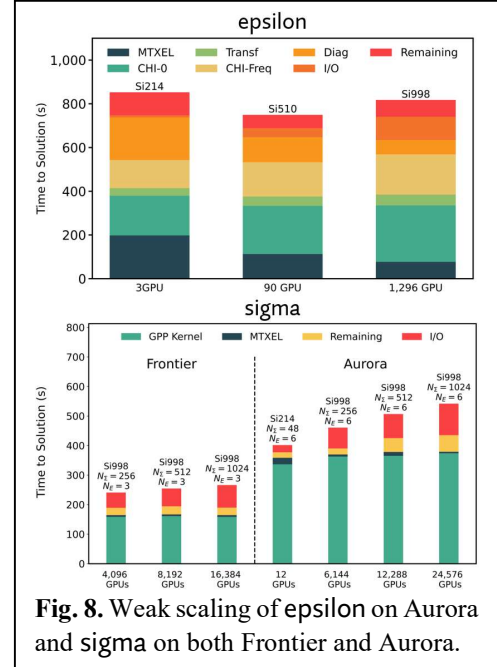


Fig. 8. Weak scaling of epsilon on Aurora and sigma on both Frontier and Aurora.

3.3. Parallel Performance

Here, we present the GPU performance results of the epsilon and sigma modules on Frontier (OLCF)

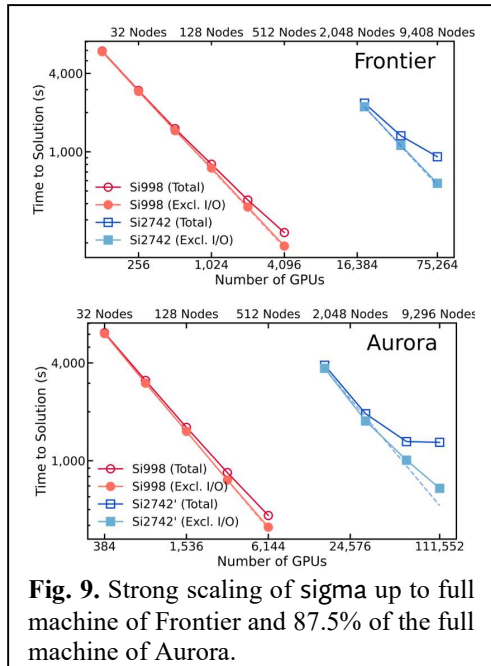


Fig. 9. Strong scaling of sigma up to full machine of Frontier and 87.5% of the full machine of Aurora.

and Aurora (ALCF). These two modules are most heavily adopted in essentially all proposed research. We present results [20] obtained for real type applications, including silicon (Si) divacancy (up to 2,742 atoms), boron nitride (BN) moiré bilayer with defect (867 atoms with vacuum), and lithium hydride (LiH) defect (up to 17,574 atoms).

Weak scaling performance: We present the weak scaling results of epsilon and sigma in Fig. 8, displaying an asymptotical $O(N^4)$ scaling. Here, the weak scaling is evaluated in terms of time to solution, where the ideal weak scaling is achieved by a constant time to solution for calculations of increasing sizes, obtained by matching the increase of computational cost with a corresponding increase in GPU resources. Excellent weak scaling is measured for both epsilon and sigma. The I/O overhead grows as the requested resources increase, which can be alleviated using node-local SSD (burst buffer) on Frontier. On Aurora, prestaging the input data set can be utilized to reduce I/O at a low cost.

Strong scaling performance: We focus on the strong scaling performance of the sigma module with the hardware-

specific optimizations in the core generalized plasmon-pole (GPP) kernel, i.e., HIP for Frontier (AMD GPUs) and SYCL for Aurora (Intel GPUs). **Fig. 9** shows the strong scaling of overall performance on Frontier and Aurora. Our results show excellent strong scaling excluding I/O, up to the full machine of Frontier with 9,408 nodes, and up to 87.5% of the full machine of Aurora with 9,296 nodes.

Peak performance: **Fig. 10** shows the throughput performance of sigma-GPP kernel (the most computationally intensive kernel) on Frontier and Aurora. On both Frontier and Aurora, the GPP kernel consistently reaches over 500 PetaFLOP/s at (nearly) the full machine scale with the hardware-optimized implementations. In particular, at full machine of Frontier (9408 nodes, or 75,264 AMD GPUs), we achieved 558.3 PetaFLOP/s in double precision, corresponding to 31.04% of the theoretical peak; with 87.5% of the full Aurora (9,296 nodes, or 111,552 Intel GPUs), we achieved 500.97 PetaFLOPs in double precision, corresponding to 39.39% of the attainable peak.

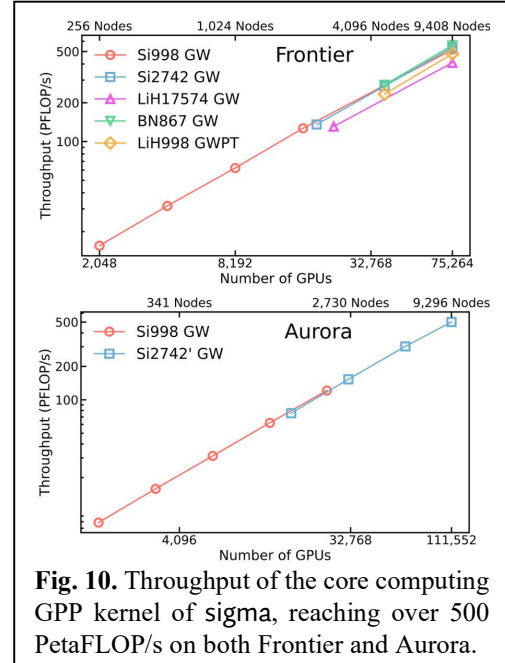


Fig. 10. Throughput of the core computing GPP kernel of sigma, reaching over 500 PetaFLOP/s on both Frontier and Aurora.

3.4. Developmental Work

We plan developmental work on several key modules of BerkeleyGW. The GWPT calculation goes through a process of computing off-diagonal matrix elements of the first-order linear-response changes in self-energy $\partial_{\mathbf{q}\nu}\Sigma$ in a similar way for the standard off-diagonal matrix elements of the self-energy. We (**Del Ben, Li**) aim to revamp the structure of the computing kernel of off-diagonal matrix elements to allow for a similar computational cost as compared with diagonal matrix elements. This restructuring of the code can be mapped into two matrix-matrix products, allowing us to leverage the highly efficient ZGEMM library calls. This new development will benefit off-diagonal calculations in GW and GWPT, potentially at a higher peak performance. The new layout will also build the basis for further developments of full-frequency treatment, self-consistent GW, and analytical gradients. We (**Jornada, Del Ben**) will also address another primary performance bottleneck in large-scale GW calculations, the construction of stochastic pseudobands, which needs the full diagonalization of the Hamiltonian. We propose to construct the pseudobands using a diagonalization-free Chebyshev-Jackson expansion, which reduces the scaling from $O(N^3)$ to $O(N^2)$ and is well-suited for GPU acceleration.

We (**Del Ben, Li, Louie, Neaton**) aim to further scale up the GW-BSE calculations, particularly for the kernel module (the computational cost of absorption is relatively lower for larger materials systems). Our latest optimization effort of the kernel module for Aurora has pushed the limit of GW-BSE calculations to an unprecedented scale for a moiré superlattice with $\sim 10,000$ atoms (**Sec. 2.4**), whereas the applications usually found in the literature are limited to few hundreds of atoms. Several challenges have been overcome in the new optimized implementation, mainly to circumnavigate the high communication cost and memory needs required for these massive applications. In particular, input data (inverse dielectric matrix) has a size of 2 TB (moiré superlattice) and needs to be iterated among MPI ranks. To address this challenge, we have crafted a tailored implementation that batches the data and recompute necessary intermediates for each batch, with small overhead but significantly reduced the communication. This strategy allows us to scale up to a few thousand Aurora nodes. We will continue to scale up the kernel module, aiming at full machine scale on both Aurora and Frontier. Moreover, we (**Del Ben, Li, Louie, Qiu**) will work on enabling the GPU acceleration of the dynamics module, which is under fast methodological development for nonequilibrium TDGW simulations. We will start with open programming models and then followed by hardware-specific optimizations, leveraging our experiences in the exascale enablement for epsilon, sigma, and kernel modules.

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Proposal Title: Exascale Simulations of Excited-State Phenomena in Quantum Materials

Year 1		Total number of node-hours for Year 1: 950,000	
Milestone:	Details (as appropriate):	Dates:	Status: (renewals only)
Development of GWPT Fröhlich interaction and applications	Resource: Frontier & Aurora Node-hours: 150,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 30 TB (Feb./26 – Apr./26) Archival storage (TB and dates): 5 TB (Mar./26 – Aug./26) Software Application: ABINIT, BerkeleyGW, EPW Tasks: Develop and implement analytical interpolation approach for Fröhlich interaction with GWPT, apply the method to polar semiconductors Dependencies: None	<i>Jan./26 – Aug./26</i>	N/A
Computation of excited-state force and structural relaxation	Resource: Aurora & Polaris Node-hours: 100,000 Production size runs (number of nodes): 4,000 Filesystem storage (TB and dates): 10 TB (Mar./26 – Aug./26) Archival storage (TB and dates): 2 TB (Mar./26 – Dec./26) Software Application: Quantum Espresso, BerkeleyGW Tasks: Compute excited-state forces from finite differences with GW-BSE and perform structural relaxations for 2D defect systems and photodissociation of water molecule on TiO ₂ Dependencies: None	<i>Mar./26 – Dec./26</i>	N/A
Computation of self-trapped excitons	Resource: Aurora & Polaris Node-hours: 50,000 Production size runs (number of nodes): 4,000 Filesystem storage (TB and dates): 5 TB (May/26 – Oct./26) Archival storage (TB and dates): 1 TB (May/26 – Dec./26) Software Application: Quantum Espresso, ABINIT, BerkeleyGW Tasks: Building excited-state forces using linear-response method based on GW-BSE, DFPT, and GWPT to study self-trapped excitons in halide perovskite Dependencies: None	<i>May/26 – Dec./26</i>	N/A
Computation of phonon screening in excitons	Resource: Frontier Node-hours: 100,000 Production size runs (number of nodes): 6,000 Filesystem storage (TB and dates): 10 TB (Jan./26 – Aug./26) Archival storage (TB and dates): 2 TB (Mar./26 – Oct./26) Software Application: Quantum Espresso, BerkeleyGW, EPW	<i>Jan./26 – Oct./26</i>	N/A

	Tasks: Computing phonon screening effects in GW-BSE for excitons of halide perovskite Dependencies: None		
Computation of Wigner crystalline excitons	Resource: Frontier & Aurora Node-hours: 140,000 Production size runs (number of nodes): 6,000 Filesystem storage (TB and dates): 50 TB (Jan./26 – Dec./26) Archival storage (TB and dates): 10 TB (Mar./26 – Dec./26) Software Application: Quantum Espresso, SIESTA, BerkeleyGW Tasks: Computing properties of WCEs in transition-metal dichalcogenides moiré superlattices using GW-BSE Dependencies: None	Jan./26 – Dec./26	N/A
Large-scale GW calculations of moiré superlattices	Resource: Frontier Node-hours: 30,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 5 TB (Apr./26 – Aug./26) Archival storage (TB and dates): 1 TB (May/26 – Oct./26) Software Application: Quantum Espresso, BerkeleyGW Tasks: Performing GW calculations on moiré superlattices with stochastic pseudobands method Dependencies: None	Apr./26 – Oct./26	N/A
Creation of (generalized) Wigner crystals in ground states	Resource: Aurora Node-hours: 10,000 Production size runs (number of nodes): 2,000 Filesystem storage (TB and dates): 2 TB (Jan./26 – Aug./26) Archival storage (TB and dates): 1 TB (Feb./26 – Oct./26) Software Application: Quantum Espresso, BerkeleyGW Tasks: Performing ground-state calculations with hybrid and other advanced functionals beyond DFT+ <i>U</i> for (generalized) Wigner crystal states Dependencies: None	Jan./26 – Oct./26	N/A
Computation of excitonic insulator phases of 2D materials	Resource: Frontier & Aurora Node-hours: 120,000 Production size runs (number of nodes): 6,000 Filesystem storage (TB and dates): 10 TB (Mar./26 – Dec./26) Archival storage (TB and dates): 2 TB (Apr./26 – Dec./26) Software Application: Quantum Espresso, ABINIT, BerkeleyGW Tasks: Computing excitonic insulators and exciton condensation in 2D materials using GW-BSE and many-body Hamiltonian Dependencies: None	Mar./26 – Dec./26	N/A

Nonequilibrium exciton-phonon coupling simulations	Resource: Frontier & Aurora Node-hours: 30,000 Production size runs (number of nodes): 3,000 Filesystem storage (TB and dates): 20 TB (May/26 – Dec./26) Archival storage (TB and dates): 5 TB (Jun./26 – Dec./26) Software Application: Quantum Espresso, ABINIT, BerkeleyGW, EPW Tasks: Computing nonequilibrium exciton-phonon dynamics in monolayer MoS ₂ using TD-aGW-ph Dependencies: “GPU enablement of TD-aGW and TD-aGW-ph”	May/26 – Dec./26	N/A
Data compression acceleration for TDGW methods	Resource: Frontier & Aurora Node-hours: 20,000 Production size runs (number of nodes): 3,000 Filesystem storage (TB and dates): 5 TB (Apr./26 – Dec./26) Archival storage (TB and dates): 2 TB (May/26 – Dec./26) Software Application: Developing SVD data compression techniques for TDGW methods Dependencies: None	Apr./26 – Dec./26	N/A
Dynamical formation processes of (excitonic) polarons	Resource: Frontier & Aurora Node-hours: 50,000 Production size runs (number of nodes): 3,000 Filesystem storage (TB and dates): 20 TB (May/26 – Dec./26) Archival storage (TB and dates): 5 TB (Jun./26 – Dec./26) Software Application: Quantum Espresso, ABINIT, BerkeleyGW, EPW Tasks: Computing the formation processes of polarons and excitonic polarons using TD-aGW-ph Dependencies: “GPU acceleration of TD-aGW and TD-aGW-ph”	May/26 – Dec./26	N/A
Exascale enablement of GW-BSE	Resource: Frontier & Aurora & Polaris Node-hours: 100,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 5 TB (Jan/26 – May/26) Archival storage (TB and dates): 2 TB (Feb./26 – Jul./26) Software Application: Quantum Espresso, ABINIT, BerkeleyGW Tasks: Scaling up the GPU implementation of GW-BSE to full-machine scale on Frontier and Aurora Dependencies: None	Jan/26 – Jul./26	N/A
GPU acceleration of TDGW methods	Resource: Frontier & Aurora & Polaris Node-hours: 50,000 Production size runs (number of nodes): 3,000 Filesystem storage (TB and dates): 2 TB (Jan/26 – May/26) Archival storage (TB and dates): 1 TB (Feb./26 – Jun./26) Software Application: BerkeleyGW	Jan/26 – Jun./26	N/A

	Tasks: Implementing GPU acceleration for TD-aGW and TD-aGW-ph methods using open programming models Dependencies: None		
Year 2		Total number of node-hours for Year 2: 950,000	
Development of long-range of quadrupolar e -ph interaction with GWPT	Resource: Frontier & Aurora Node-hours: 150,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 20 TB (Jan./27 – Dec./27) Archival storage (TB and dates): 5 TB (Jan./27 – Dec./27) Software Application: ABINIT, BerkeleyGW, EPW Tasks: Develop and implement analytical interpolation approach quadrupolar interaction with GWPT, apply the method to bulk and 2D materials Dependencies: “Development of GWPT Fröhlich interaction and applications”	<i>Jan./27 – Dec./27</i>	N/A
Applications of excited-state forces in single-photon emitter and photodissociation pathways	Resource: Aurora & Frontier Node-hours: 150,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 20 TB (Jan./27 – Oct./27) Archival storage (TB and dates): 5 TB (Jan./27 – Dec./27) Software Application: Quantum Espresso, BerkeleyGW Tasks: Compute properties and single-photon emitter and various pathways for photodissociation with excited-state forces Dependencies: “Computation of excited-state force and structural relaxation”, “Exascale enablement of GW-BSE”	<i>Jan./27 – Dec./27</i>	N/A
Phonon screening of excitons including polaronic effects and exciton Wannier functions	Resource: Aurora Node-hours: 100,000 Production size runs (number of nodes): 6,000 Filesystem storage (TB and dates): 5 TB (Jan./27 – Aug./27) Archival storage (TB and dates): 1 TB (Jan./27 – Oct./27) Software Application: Quantum Espresso, ABINIT, BerkeleyGW Tasks: Incorporating polaronic effects in phonon screening of excitons, with analysis using exciton Wannier functions Dependencies: “Computation of phonon screening in excitons”, “Exascale enablement of GW-BSE”	<i>Jan./27 – Oct./27</i>	N/A
Exascale computation of Wigner crystalline excitons	Resource: Frontier & Aurora Node-hours: 120,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 50 TB (Jan./27 – Dec./27) Archival storage (TB and dates): 10 TB (Mar./27 – Dec./27) Software Application: Quantum Espresso, SIESTA, BerkeleyGW Tasks: Performing exascale GW-BSE calculations for WCEs	<i>Jan./27 – Dec./27</i>	N/A

	Dependencies: “Computation of Wigner crystalline excitons”, “Exascale enablement of GW-BSE”		
Exascale GW calculations of complicated moiré heterostructures	Resource: Frontier Node-hours: 60,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 5 TB (Jan./27 – Jun./27) Archival storage (TB and dates): 1 TB (Feb./27 – Aug./26) Software Application: Quantum Espresso, BerkeleyGW Tasks: Performing exascale GW calculations on moiré heterostructures complicated multilayer stacking and twisting Dependencies: “Large-scale GW calculations of moiré superlattices”	<i>Jan./27 – Aug./27</i>	N/A
Exascale computation of excitonic insulator phases	Resource: Frontier & Aurora Node-hours: 120,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 10 TB (Jan./27 – Aug./27) Archival storage (TB and dates): 2 TB (Jan./27 – Oct./27) Software Application: Quantum Espresso, ABINIT, BerkeleyGW Tasks: Computing excitonic insulators and exciton condensation in 2D materials using GW-BSE and many-body Hamiltonian Dependencies: “Computation of excitonic insulator phases of 2D materials”, “Exascale enablement of GW-BSE”	<i>Jan./27 – Oct./27</i>	N/A
Large-scale nonequilibrium dynamics simulations	Resource: Frontier & Aurora Node-hours: 50,000 Production size runs (number of nodes): 6,000 Filesystem storage (TB and dates): 20 TB (Jan./27 – Dec./27) Archival storage (TB and dates): 10 TB (Jan./27 – Dec./27) Software Application: Quantum Espresso, ABINIT, BerkeleyGW, EPW Tasks: Large-scale simulations of nonequilibrium dynamical phenomena with GPU and data-compression acceleration Dependencies: “GPU enablement of TD-aGW and TD-aGW-ph”, “Nonequilibrium exciton-phonon coupling simulations”, “Exascale enablement of GW-BSE”	<i>Jan./27 – Dec./27</i>	N/A
Large-scale simulations of (excitonic) polaron dynamics	Resource: Frontier & Aurora Node-hours: 50,000 Production size runs (number of nodes): 6,000 Filesystem storage (TB and dates): 20 TB (Jan./27 – Dec./27) Archival storage (TB and dates): 10 TB (Jan./27 – Dec./27) Software Application: Quantum Espresso, ABINIT, BerkeleyGW, EPW Tasks: Large-scale simulations of (excitonic) polaron dynamics of selected materials systems	<i>Jan./27 – Dec./27</i>	N/A

	Dependencies: “GPU acceleration of TD-aGW and TD-aGW-ph”, “Dynamical formation processes of (excitonic) polarons”, “Exascale enablement of GW-BSE”		
Exascale optimization and calculations	Resource: Frontier & Aurora Node-hours: 150,000 Production size runs (number of nodes): 9,000 Filesystem storage (TB and dates): 10 TB (Jan./27 – Aug./27) Archival storage (TB and dates): 5 TB (Feb./27 – Nov./27) Software Application: Quantum Espresso, ABINIT, BerkeleyGW Tasks: GPU architecture-specific optimization of GW-BSE and TDGW approaches for exascale calculations Dependencies: “Exascale enablement of GW-BSE”, “GPU acceleration of TDGW methods”	<i>Jan./27 – Nov./27</i>	N/A